



Real-time Deployment

LECTURE

Introduction to Real-time Deployment



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In this lecture Introduction to Real-time Deployment, we discuss real-time deployment, challenges of building real-time AI systems, and Databricks Model Serving.

Real-time Deployment

- The process of serving machine learning models in a production environment where **predictions are generated instantly** in response to incoming data or requests.
- Crucial for applications that require **low-latency responses**, such as chatbots, message intent detection, autonomous systems, and other time-sensitive tasks.

With the emergence of new Gen AI applications, this deployment method is becoming increasingly common, especially as large language models need to be served in real-time.



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First, let's go over the basics of real-time deployment. This refers to serving a machine learning or language model in a production environment where predictions are generated instantly. The goal is to provide quick access to completions, which is crucial for low-latency use cases like chatbots or message intent detection. For example, if a user posts a message on a social media platform, you might want to immediately check that it doesn't contain violent content. As mentioned earlier, with the rise of new generative AI applications and the complexity of running batch jobs on your own infrastructure, real-time deployment is becoming more and more common—especially since large language models are often expected to deliver results instantly.

Real-time Deployment

A typical real-time deployment **use case**

- **Description:** Real-time Intent Detection for social media platform
- **Scenario:**
 - A fine-tuned large language model for classifying social media post, is deployed in the real-time environment.
 - The model provides immediate classification of posts over a REST API.
 - If post content is classified as toxic/violent/harmful it is taken down/removed.
- **Requirements:** Low latency, immediate action, 24/7 uptime, continuous monitoring

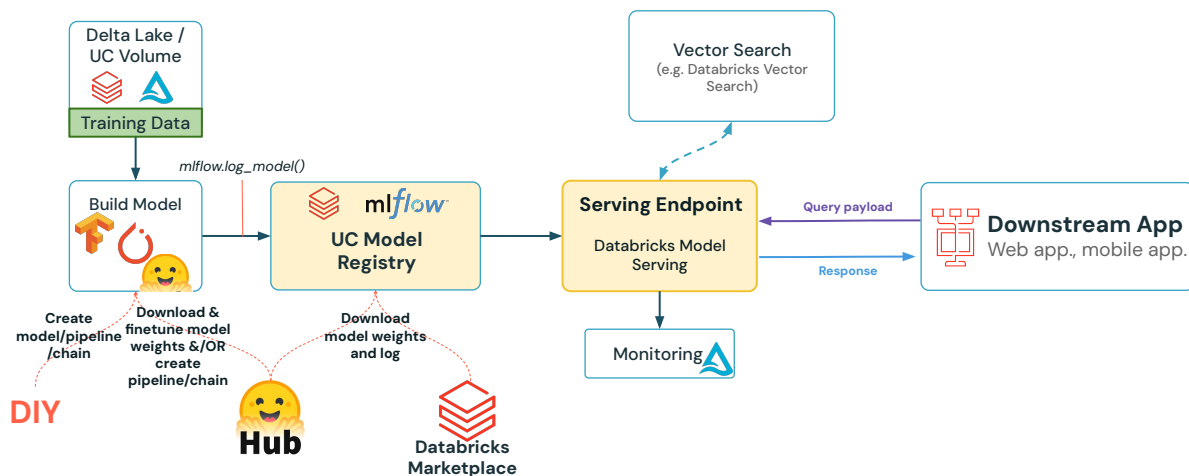


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To give an example, a typical real-time deployment scenario would be intent detection on a social media platform. In this case, you might deploy a fine-tuned large language model or use a proprietary model with few-shot classification capabilities. The goal is for the model to immediately classify user posts as they're submitted to the platform. Through a REST API, if a post is flagged as harmful, toxic, or violent, it should be taken down right away. This kind of setup demands low latency, instant action, and round-the-clock uptime with ongoing monitoring.

Real-time Deployment

A typical real-time model deployment workflow for LMs



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Going back to how this affects development and deployment in Databricks, the approach to developing the model itself doesn't really change. Whether you're using a model from the marketplace, fine-tuning a model from the model hub or Databricks' supported foundation models with the fine-tuning API, or even building your own model from scratch, the development process remains consistent. Once the model is registered, deploying it to a model serving endpoint is straightforward and efficient. As queries come in from downstream applications—like chatbots or web applications—the serving endpoint delivers responses in real time. If you're building a retrieval-augmented generation (RAG) application, vector search can be utilized as well. Throughout this process, everything happening at the endpoint, from hardware utilization to each request and response, is automatically logged and monitored.

Challenges with building Real-time AI Systems

Most AI systems don't get into production



Infrastructure is hard

Real-time AI systems require fast and scalable serving infrastructure, which is costly to build and maintain



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So, what are the challenges in building these kinds of systems? Even though it may sound straightforward, infrastructure can be quite complicated. It's not easy to access GPUs and serve either multiple models or a single model to many users across different regions while also making sure hardware is used efficiently. The main issue is that this hardware is expensive, so getting the most out of what's provisioned is crucial. This is where REST or real-time APIs come in—using them can help ease some of the burden, at least from the consumer's perspective.

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Deploying real time models needs disparate tools

Data teams use diverse tools to develop models

Customers use separate platforms for data, LLMs and Serving, adding complexity and cost



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If you've ever attempted to set this up on your own, you've probably noticed that it involves working with several different stacks and tools at once, and managing to keep everything stable, connected, and in sync can be pretty challenging.

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Operating production AI requires expert resources

Steep learning curve of deployment tools.

Model deployment is bottlenecked by limited engineering resources, limiting the ability to scale AI



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On top of that, there's the ongoing operational effort and engineering hours needed to keep everything running smoothly. Someone has to be ready to step in if anything breaks, plus it's important to always use the latest and best technology in the serving stack. All of these resources and expertise aren't readily available to most individuals who might want to harness the power of generative AI.

Databricks Model Serving

Integrate your model into your websites and applications as an API



Production-Grade Serving

Highly available, low latency, scalable serving that works for small and large workloads



Accelerate deployments with Lakehouse-Unified Serving

Automatic feature lookups, monitoring and unified governance that **automates deployment** and reduce errors



Simplified Deployment

Simple and flexible deployment through **UI** or **API**



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Finally, Databricks Model Serving streamlines deployment with its user-friendly and flexible UI or API. Creating model endpoints is made easy with just a few clicks, allowing you to swiftly deploy your models.



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